THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 9: **Propositions and their Converses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. For the expression $n^2 + n + 107$, the smallest natural number n for which the value is NOT prime is: 1

(a) 4 **(b)** 2 **(c)** 3 **(d)** 5

2. The number of factors of 180 is: 1

(a) 17 **(b)** 20 **(c)** 19 **(d)** 18

3. A triangle has sides 20 cm, 99 cm and 101 cm. Using the converse of the Baudhāyana–Pythagoras theorem, the triangle is: 1

(a) equilateral **(b)** right-angled **(c)** isosceles **(d)** not right-angled

4. Which statement is true for every integer n? 1

(a) $n^2 + n$ is odd **(b)** $n^2 - 1$ is even **(c)** $n^2 + n$ is even **(d)** $2n + 1$ is even

5. The sum of the 7 consecutive integers starting from 23 is: 1

(a) 182 **(b)** 183 **(c)** 175 **(d)** 189

6. The number of integers from 1 to 130 that have an odd number of factors is: 1

(a) 10 **(b)** 12 **(c)** 13 **(d)** 11

7. The converse of "If a triangle is equilateral, then it is isosceles" is false because: 1

(a) a triangle with sides $5, 5, 8$ is isosceles but not equilateral

(b) two angles of 60° in an equilateral triangle are equal but not vertically opposite

(c) $x = -3$ gives $x^2 = 9$ but $x \neq 3$

(d) a 4 cm by 6 cm rectangle is not a square

8. Which of the following is a counterexample to the statement "$2n + 1$ is prime for every natural number n"? 1

(a) $n = 4$ **(b)** $n = 2$ **(c)** $n = 3$ **(d)** $n = 1$

9. For the expression $n^2 - n + 29$, the smallest natural number n for which the value is NOT prime is: 1

(a) 4 **(b)** 2 **(c)** 5 **(d)** 3

10. The number of factors of 144 is: 1

(a) 15 **(b)** 14 **(c)** 16 **(d)** 17

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the smallest natural number n for which $n^2 + n + 11$ is not prime, and the value obtained. 2

12. Is the number 150 a perfect square? Decide by counting its factors. 2

13. Write the converse of "If two angles are vertically opposite, then the two angles are equal" and test it. 2

14. Find the smallest natural number n for which $n^2 - n + 3$ is not prime, and the value obtained. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the converse of "If a number is a multiple of 15, then it is a multiple of 5". Is the converse true? Justify. 3

16. Show that the statement "$n^2 + n + 41$ is prime for every natural number n" is false by giving a counterexample. 3

17. If n is a multiple of 30, is n a multiple of 10? Write the converse and give a counterexample if false. 3

OR

Write the converse of "If a number is divisible by 12, then it is divisible by 4". Is the converse true? Justify. 3

18. Show that the statement "The difference of two primes is prime" is false by giving a counterexample. 3

OR

If n is a multiple of 36, is n a multiple of 9? Write the converse and give a counterexample if false. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that the sum of 7 consecutive even numbers starting at $2n$ is a multiple of 14. Check with $n = 3$. 5

OR

Take the two-digit number 85. Subtract the sum of its digits and show that the result is divisible by 9. Prove the same for any two-digit number. 5

20. Prove that the sum of 3 consecutive even numbers starting at $2n$ is a multiple of 6. Check with $n = 10$. 5

OR

Take the two-digit number 92. Subtract the sum of its digits and show that the result is divisible by 9. Prove the same for any two-digit number. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND